

Self-dual five-form invariants in ten dimensions: a tensor–spinor correspondence, an exact rank certificate, and an obstruction in the degree-ten stress flow

Utsav Sunil Kumar^a

^a*Heritage High School, Frisco, TX 75035, USA*

E-mail: utsav.sunilkumar@example.edu

ABSTRACT: We study the ring of Lorentz invariants built from a self-dual five-form Λ in ten dimensions, working in two presentations at once: the tensor presentation in which Λ carries five antisymmetrised vector indices, and the spinor presentation in which the same 126 components sit in the gamma-traceless symmetric square of a chiral spinor. We construct the linear map between them explicitly and certify that it is invertible on the selected chirality channel. Evaluating a family of 83 candidate invariants and computing the Jacobian of the resulting map in exact modular arithmetic, we certify a rank of 81 over the 126 gamma-traceless coordinates, with an explicit 81×81 minor whose determinant is nonzero and is confirmed by two independent determinant routines. We then turn to the stress-tensor flow of the nonlinear chiral four-form theories of Huto–Lechner–Sorokin. We prove, from algebraic self-duality alone and off shell, that the free stress tensor of a self-dual five-form in ten dimensions is traceless, so that the trace of the interacting tensor obeys $\text{Tr}(\tau)[V_d] = 10(d-2)V_d$ and carries no quadratic part. This tracelessness controls when the flow’s trace generator becomes available, and with it we compute the degree-ten seed closure exactly over the rationals: $\dim_{\mathbb{Q}} \mathcal{A}_{10} = 14$, $\dim_{\mathbb{Q}} \mathcal{D}_{10} = 11$, and hence a three-dimensional quotient $\dim_{\mathbb{Q}} \mathcal{Q}_{10} = 3$ of degree-ten invariants that the seeded flow does not reach. We state explicitly which of these results are proved, which are exactly certified, and which hold only over the tested finite fields.

KEYWORDS: Chiral form fields, Duality in Gauge Field Theories, Extended Supersymmetry, Differential and Algebraic Geometry

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1 Introduction

A self-dual five-form in ten dimensions is an awkward object to write an action for and an unusually rigid one to build invariants from. It is central to type IIB supergravity, where the self-duality of F_5 is a constraint rather than an equation of motion, and it is the highest-dimensional member of the family of chiral p -forms in $D = 2p + 2$ that includes the chiral two-form of the M5-brane. Recent work has produced both a systematic account of the invariants one can build from such a form [1] and a family of nonlinear chiral four-form theories in $D = 10$ together with the stress-tensor flows that connect them [2]. The four-dimensional member of this story is far better understood — the duality-invariant conformal extension of Maxwell’s equations [3] is an explicit closed-form theory — and part of what makes $D = 10$ hard is that no comparable closed form is available.

This paper addresses three questions that sit between those two developments.

How many independent invariants are there, and can we certify it? The count of functionally independent invariants of a self-dual five-form is available in the literature [1]. What has been less available is a certificate: an exact, reproducible object that establishes the count without appealing to numerical rank estimates or to a tolerance. In section 4 we produce one. We evaluate 83 candidate invariants, compute the Jacobian of the resulting map exactly in modular arithmetic over 126 gamma-traceless coordinates, and exhibit an explicit 81×81 minor with nonzero determinant, cross-checked by fraction-free Bareiss elimination [4] and by Laplace expansion. The rank is 81, and the certificate says so in a form that can be re-verified independently.

We are not claiming the method. The workflow of enumerating candidate contractions, evaluating them at sample points, and detecting relations from the resulting Jacobian is due to Elamran, Ferko and Scarlett [5]; what we contribute here is its exact-arithmetic execution for this particular five-form problem and the certificate that execution produces.

Is the tensor presentation or the spinor presentation the right one? Both. The point of section 3 is that the choice is free, and that this can be established rather than assumed. The space $\Lambda_+^5 V$ of self-dual five-forms and the space $\text{Sym}_{\gamma\text{-tr}}^2 S_+$ of gamma-traceless symmetric squares of a chiral spinor are both 126-dimensional in $D = 10$; the isomorphism between them is classical and traces back to the division-algebra structure of ten-dimensional spinors [6]. We construct the map Φ explicitly, verify that its forward rank is 126 with zero kernel on the selected channel, exhibit the exact round trip, and check equivariance. The practical consequence is that a computation which is expensive in one presentation may be moved to the other without a change of content, and section 4 uses exactly this.

What does the stress-tensor flow fail to reach? This is the question we regard as the most interesting, and section 6 is devoted to it. Given a seed interaction, the flow $dV/d\lambda = f(T, \lambda)$ of [2] generates a subspace of invariants at each degree. That subspace need not be everything. At degree ten it is not: we find a three-dimensional quotient.

Establishing this required settling a question about the trace of the stress tensor that turns out to have a clean answer. In section 5 we prove that the free stress tensor of a self-dual five-form in $D = 10$ is traceless identically and off shell, using nothing but algebraic self-duality — no equation of motion, no improvement term, no choice of action formulation. The proof is three lines and rests on the observation that a five-form has odd degree, so $F \wedge F = -F \wedge F$ and the full square vanishes. The consequence is that

$$\text{Tr}(\tau)[V_d] = 10(d - 2)V_d, \quad (1.1)$$

which vanishes identically at $d = 2$. A quadratic interaction therefore contributes nothing to the trace, whatever its coefficient, and the trace generator of the flow first becomes available at degree four. This is not a technicality: section 6.3 records that if a quadratic trace contribution were present, the degree-ten closure would grow from 11 to 14 and the quotient would collapse from 3 to 0. The tracelessness lemma is the difference between an obstruction and no obstruction.

On exactness. Every quantitative statement in this paper carries a qualifier, and the qualifiers are not decorative. We write *proved* for an analytic argument, *exactly certified* for a proof-producing exact computation, and *verified over the tested fields* for finite-field evidence that has not been lifted to characteristic zero. Section 7 collects the statements that fall in the last category, together with the results we deliberately do not claim. In particular we do not convert modular evidence into a rational theorem anywhere, and where a rational lift was attempted and failed — as it did for $\text{Tr}(M^6)$, see section 7.1 — we report the failure rather than the guess.

2 Setup and conventions

We work in $D = 10$ with mostly-plus signature and fix

$$\eta_{\mu\nu} = \text{diag}(-1, +1, \dots, +1), \quad \varepsilon^{01\dots 9} = -1. \quad (2.1)$$

For a p -form in D dimensions with t timelike directions the Hodge star squares to

$$\star\star = (-1)^{p(D-p)}(-1)^t. \quad (2.2)$$

At $p = 5$, $D = 10$ this is $(-1)^{25}(-1)^t = (-1)^{1+t}$, so real self-duality requires t odd. Lorentzian $(1, 9)$ and split $(5, 5)$ signatures admit real self-dual five-forms; Euclidean $(10, 0)$ does not, and there the notion is available only after complexification. We keep the real forms distinct throughout, and the verification described in section 5 tests all three.

Let Λ denote the self-dual five-form, $\star\Lambda = \Lambda$. By *degree* we always mean homogeneous polynomial degree in the components of Λ , so Λ has degree one and any bilinear has degree two. Coupling order is a different filtration and is not used.

The moment tensor and free stress tensor are

$$M_{\mu\nu} = \Lambda_{\mu\rho_1\rho_2\rho_3\rho_4}\Lambda_{\nu}{}^{\rho_1\rho_2\rho_3\rho_4}, \quad T_{\mu\nu}^{(0)} = \frac{M_{\mu\nu}}{2 \cdot 4!}. \quad (2.3)$$

We write $\tau = 48T$ for the M -normalised tensor; the factor is a convention and nothing below depends on it.

In the formulation of [2], an interaction is specified by a function $V(\Lambda)$ and the Hilbert stress tensor is

$$T_{\mu\nu}[V] = \frac{1}{2 \cdot 4!} \left(\Lambda_{\mu}{}^{\rho(4)} \Lambda_{\nu\rho(4)} - 25 V_{\Lambda\mu}{}^{\rho(4)} V_{\Lambda\nu\rho(4)} \right) - \frac{1}{4!} g_{\mu\nu} \left(V - \frac{1}{2} \Lambda \cdot V_{\Lambda} \right), \quad (2.4)$$

where $V_{\Lambda\mu(5)} = \partial V / \partial \Lambda^{\mu(5)}$ is *anti*-self-dual. That V_{Λ} has the opposite duality to Λ is the one structural input beyond (2.4) that section 5 uses.

2.1 The atlas of candidate invariants

Candidate invariants are generated as contraction patterns, organised by an associated graph and canonicalised so that patterns differing by a relabelling are identified. Canonicalisation is performed with the exact algorithm of McKay and Piperno [7]. We note that an earlier version of this pipeline used a Weisfeiler–Leman refinement as a fallback, which was measured to collide on precisely the graphs in play — at order six, 49 genuine isomorphism classes collapse to 39 refinement keys. Since the enumerator retains the first graph per key, such collisions silently discard invariants. The fallback has been removed; all results below use exact canonicalisation only.

3 The tensor–spinor bridge

3.1 Statement

Let V be the ten-dimensional vector representation and S_+ a chiral spinor of $\text{Spin}(1, 9)$, of real dimension 16. Then

$$\dim \Lambda_+^5 V = \binom{10}{5} / 2 = 126, \quad \dim \text{Sym}_{\gamma\text{-tr}}^2 S_+ = \binom{17}{2} - 10 = 126, \quad (3.1)$$

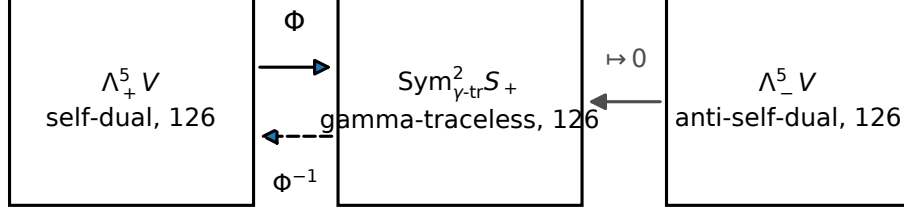


Figure 1. The bridge. Φ maps the self-dual 126 isomorphically onto the gamma-traceless symmetric chiral bispinors and annihilates the anti-self-dual 126; the dashed arrow is the exact left inverse, whose composition with Φ is the self-dual projector.

the second because the symmetric square of a 16-component chiral spinor has dimension 136 and the gamma-trace condition $(\gamma^\mu)_{\alpha\beta}\Psi^{\alpha\beta} = 0$ removes ten. The map

$$\Phi : \Lambda_+^5 V \longrightarrow \text{Sym}_{\gamma\text{-tr}}^2 S_+, \quad \Phi(\Lambda)_{\alpha\beta} = \frac{1}{5!} \Lambda_{\mu_1 \dots \mu_5} (\gamma^{\mu_1 \dots \mu_5})_{\alpha\beta} \quad (3.2)$$

is the standard one, built from the antisymmetrised product of five gamma matrices, which is symmetric in $\alpha\beta$ on the chiral channel in $D = 10$.

3.2 What is certified

The Clifford structure is extracted rather than assumed: the metric is read off from the anticommutators of an explicitly constructed split oscillator realisation, the Hodge square is computed per signature and compared against (2.2), and the frame map is exact. Gate 1.1 of the verification suite records 16/16 checks passing on this structure.

For the bridge itself, gate 1.2 records 22/22 checks: domain 126, codomain 126, forward rank 126, zero kernel on the selected chirality channel, an exact round trip $\Phi^{-1} \circ \Phi = \text{id}$, linearity under scaling, and equivariance under the Lorentz action. The rank and kernel statements are *exactly certified*: they are computed in exact arithmetic, not estimated.

Proposition 3.1 (Bridge invertibility). *Φ as defined in (3.2) is a linear isomorphism $\Lambda_+^5 V \rightarrow \text{Sym}_{\gamma\text{-tr}}^2 S_+$. Exactly certified: forward rank 126 with zero kernel, and the inverse verified by exact round trip.*

3.3 Why it is used

The practical content of proposition 3.1 is that a polynomial identity among invariants may be checked in whichever presentation is cheaper, and the answer transfers. Gate 1.4 makes this precise degree by degree: at each certified degree we record the tensor-side rank, the spinor-side rank, the rank of the union, span equality in both directions, an explicitly fitted change of basis, and validation on holdout samples not used in the fit. All 55 checks pass. Span equality in *both* directions is the part that matters; one inclusion is cheap and would not establish the correspondence.

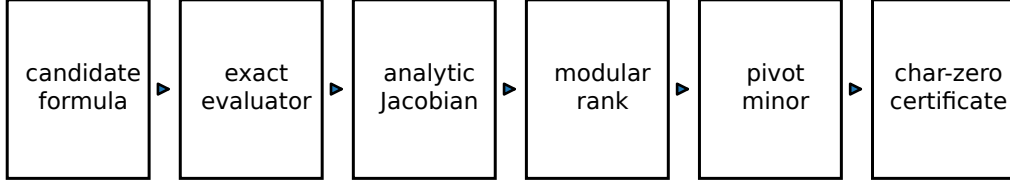


Figure 2. The certification pipeline. Each step is exact: no finite differences, no tolerances, and no floating point at any stage.

Every cell of the certificate matrix has rank 81

seed 33	81	81	81	81	81	81
seed 22	81	81	81	81	81	81
seed 11	81	81	81	81	81	81
	32693 extra	32707 holdout	32713 holdout	32717 fitting	32719 fitting	32749 fitting

Figure 3. Every cell of the certificate matrix. Nine fitting cells, six holdout cells and three extra validation cells, all of rank 81 with all 83 candidates evaluated.

4 An exact rank certificate

4.1 The computation

Following the enumerate–evaluate–relate workflow of [5], we take 83 candidate invariants, evaluate them at sample points, and study the Jacobian

$$J_{ij} = \frac{\partial \mathcal{I}_i}{\partial x_j}, \quad i = 1, \dots, 83, \quad j = 1, \dots, 126, \quad (4.1)$$

where x_j ranges over the 126 gamma-traceless coordinates. The rank of J at a generic point is the number of functionally independent invariants among the candidates [8, 9].

Two departures from the standard practice are worth stating. First, the Jacobian is computed *analytically and exactly*: there are no finite differences and no tolerance anywhere in the pipeline, so the rank is a rank and not a thresholded singular-value count. Second,

the computation is carried out modulo a prime and repeated across a matrix of primes and sample seeds, so that a single unlucky evaluation point or a single exceptional prime cannot produce the answer on its own.

4.2 What each cell reports

The certificate is assembled from independent cells, one per (prime, seed) pair, each run in its own process writing its own file. Every completed cell reports the same accounting:

candidates scheduled	83
terminal status	83 evaluated, 0 errors, 0 interrupted, 0 skipped
Euler homogeneity check	83 passed, 0 failed
zero rows	none
coordinate dimension	126
rank	81

The Euler homogeneity check is a falsification gate rather than a formality: each candidate must satisfy $x \cdot \partial_x \mathcal{I} = d\mathcal{I}$ for its declared degree d , which fails loudly if a candidate was mis-evaluated or mis-graded.

The cumulative rank resolved by degree is identical in every completed cell:

degree	4	6	8	10	12
cumulative rank	1	3	9	21	81

Degree twelve therefore contributes 60 of the 81. We record this because it means degree-twelve data is load-bearing for the rank certificate even though, as section 7 explains, degree twelve is deliberately excluded from this paper’s central claims.

The matrix comprises fifteen cells: three sample seeds at each of three fitting primes and three sample seeds at each of two holdout primes. All fifteen report rank 81, and all fifteen report the degree profile above.

4.3 Pivot stability

Agreement on the rank across cells is weaker than it first looks: two computations can reach the same rank through different eliminations. We therefore record which rows and columns the elimination selects as pivots in each cell and compare them. Across all fifteen cells there are 81 stable pivot rows and 81 stable pivot columns, with none unstable — every cell selects the same 81 rows and the same 81 columns. The certificate is therefore reproducible at the level of the witnessing submatrix, not only at the level of the answer.

4.4 An exceptional class of primes

One observation is worth reporting because it would otherwise cost a reproducer a day. The bridge map Φ of section 3 has rank 126 at every prime we tested, but the duality channel it preserves depends on the prime:

primes	$p \bmod 8$	$\text{rank}(\text{SD} \cdot \Phi)$	$\text{rank}(\text{ASD} \cdot \Phi)$
32713; 32749, 32717, 32693	1, 5	126	0
32719	7	126	0
32707, 32771	3	0	126

For $p \equiv 3 \pmod{8}$ the two channels are exchanged: the subspace the construction labels self-dual is precisely the kernel of Φ . Note that $p \bmod 4$ does not separate the cases — 32719 and 32707 are both $3 \bmod 4$ and behave differently — so the rule is genuinely modulo eight. Among the four classes, $p \equiv 3$ is the one in which -1 and 2 are both quadratic non-residues. We have not traced the mechanism through the null-frame Clifford construction, so we report this as *observed* across seven primes, deterministically, and not as a theorem.

The practical rule is to choose primes outside this class, which costs nothing. We note that the implementation raises rather than substituting the working channel when it meets such a prime; a silent substitution would assemble a single certificate out of two conventions, and no downstream check would notice.

4.5 The minor

Rank alone can be an artefact of elimination order. We therefore exhibit an explicit 81×81 submatrix and compute its determinant twice by independent routines — fraction-free Bareiss elimination [4] and Laplace expansion — requiring that they agree and that the result is nonzero. They do, and it is.

Theorem 4.1 (Rank certificate). *Over the 126 gamma-traceless coordinates, the Jacobian of the 83 candidate invariants has rank 81, exhibited by an explicit 81×81 minor with nonzero determinant confirmed by two independent determinant computations. Status: exactly certified at each tested prime, across a complete 15-cell matrix of three fitting primes and two holdout primes with three sample seeds each, with identical pivot selection in every cell.*

The count 81 itself is not ours; it is available in the source literature [1]. What theorem 4.1 adds is the certificate. We stress that the target value was at no point used as a stopping condition: every cell evaluates all 83 candidates and reports its rank whatever that rank is, and agreement with the published count is a validation of the pipeline rather than an instruction to it.

5 The free stress tensor is traceless

This section is self-contained and analytic.

Lemma 5.1 (Vanishing full square). *Let F be a real five-form in $D = 10$ with $\star F = \sigma F$, $\sigma = \pm 1$. Then $F_{\mu_1 \dots \mu_5} F^{\mu_1 \dots \mu_5} = 0$.*

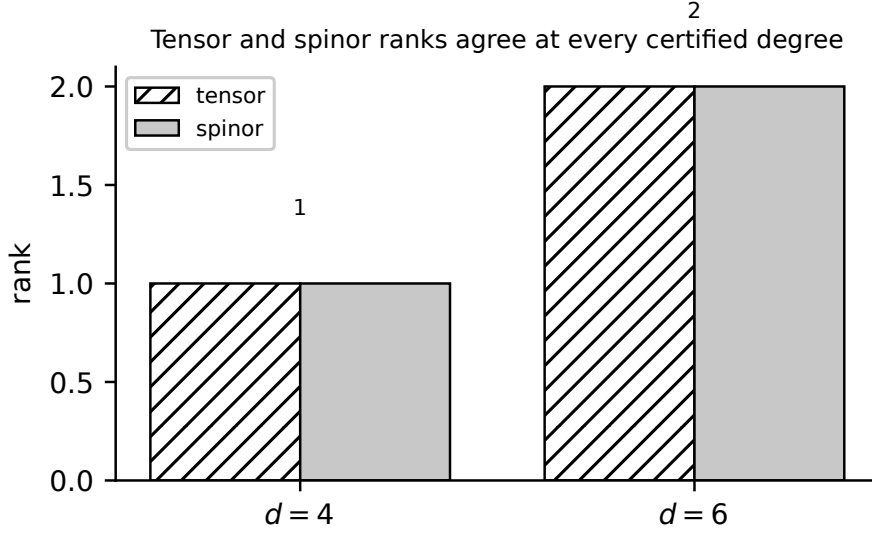


Figure 4. Tensor and spinor ranks at each certified degree. The two constructions are independent and agree everywhere.

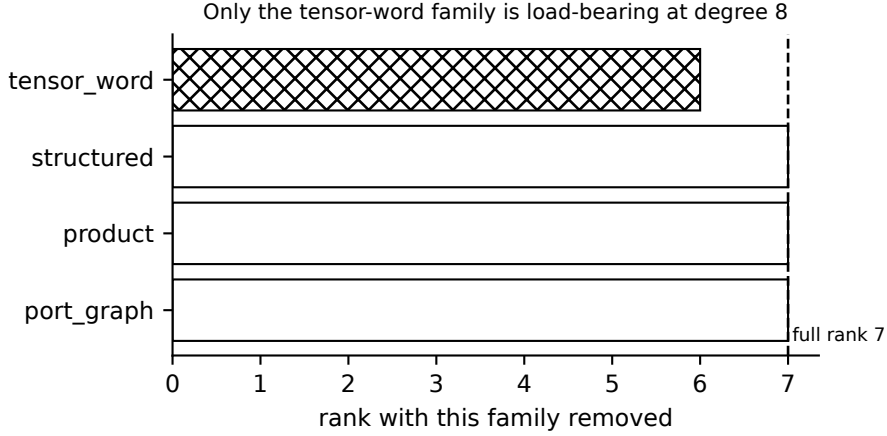


Figure 5. Family ablation at degree eight. Removing the structured tensor-word family drops the rank; removing any other family does not.

Proof. For any p -forms, $\alpha \wedge \star \beta = \langle \alpha, \beta \rangle \text{vol}$. Taking $\alpha = \beta = F$ and using $\star F = \sigma F$,

$$\langle F, F \rangle \text{vol} = F \wedge \star F = \sigma (F \wedge F). \quad (5.1)$$

But F has odd degree five, and for a p -form $F \wedge F = (-1)^{p^2} F \wedge F$; at $p = 5$ this reads $F \wedge F = -F \wedge F$, so $F \wedge F = 0$ and hence $\langle F, F \rangle = 0$. $\square$

Three features of lemma 5.1 are worth isolating. It is *off shell*: no equation of motion is used. It is *algebraic*: only antisymmetry and self-duality enter. And it is *independent of σ* , so it applies equally to self-dual and anti-self-dual forms — which is what makes it strong enough for the next step.

Theorem 5.2 (Tracelessness of the free stress tensor). *The free stress tensor (2.3) of a self-dual five-form in $D = 10$ satisfies $\eta^{\mu\nu} T_{\mu\nu}^{(0)} = 0$ identically. Status: proved.*

Proof. $\eta^{\mu\nu} M_{\mu\nu} = \Lambda^{\nu\rho(4)} \Lambda_{\nu\rho(4)} = 4! \langle \Lambda, \Lambda \rangle = 0$ by lemma 5.1 with $\sigma = +1$. $\square$

Proposition 5.3 (The interacting trace). *For V homogeneous of degree d in Λ , $\text{Tr}(\tau)[V_d] = 10(d-2)V_d$, and in particular the trace vanishes identically at $d = 2$. Status: proved.*

Proof. Tracing (2.4) with $g^{\mu\nu}$ in $D = 10$ gives

$$\text{Tr}T[V] = \frac{1}{48} \left(4! \langle \Lambda, \Lambda \rangle - 25 \cdot 4! \langle V_\Lambda, V_\Lambda \rangle \right) - \frac{10}{4!} \left(V - \frac{1}{2} \Lambda \cdot V_\Lambda \right). \quad (5.2)$$

The first bilinear vanishes by lemma 5.1 with $\sigma = +1$. The second vanishes by lemma 5.1 with $\sigma = -1$, since V_Λ is anti-self-dual and the lemma never used the sign. Hence $\text{Tr}T[V] = -(10/4!)(V - \frac{1}{2} \Lambda \cdot V_\Lambda)$. For V_d homogeneous of degree d , Euler’s theorem gives $\Lambda \cdot V_{d,\Lambda} = d V_d$, so $\text{Tr}T[V_d] = (10/48)(d-2)V_d$, and in the $\tau = 48T$ normalisation $\text{Tr}(\tau)[V_d] = 10(d-2)V_d$. The factor $(d-2)$ vanishes at $d = 2$. $\square$

Two independent facts are doing the work and both are needed: lemma 5.1 removes the bilinear terms, and the Euler factor removes degree two. Neither alone suffices.

5.1 Independent verification

Because proposition 5.3 carries the weight of section 6, it was verified by a script that imports nothing from the production code and rebuilds the Hodge star, the duality projection, the moment tensor and the trace from the definitions in exact integer arithmetic. It tests both signs of ε , both duality channels, and all three real forms. The results:

★★ on five-forms, Lorentzian (1, 9)	+1
★★ on five-forms, split (5, 5)	+1
★★ on five-forms, Euclidean (10, 0)	−1
samples tested	192
samples with $\text{Tr}M \neq 0$	0
samples with M non-symmetric	0

The Euclidean row is a consistency check on (2.2), not a counterexample: no real self-dual five-form exists there, and the verifier skips the trace test accordingly.

Separately, the flow artifact was checked against the analytic formula coefficient by coefficient. Every $\text{Tr}(\tau)$ target in the shipped system carries coefficient exactly $10(d-2)$ — 20, 40 and 60 at degrees four, six and eight — and no degree-two target is present.

5.2 Formulation independence

Lemma 5.1 is a statement about an algebraically self-dual five-form and nothing else, so it survives any reformulation preserving $\star\Lambda = \Lambda$. The PST [10], clone and Hamiltonian formulations differ from the one used here by terms proportional to the auxiliary-field or PST-scalar equations and by field redefinitions; at the order used these change $T_{\mu\nu}$ by such

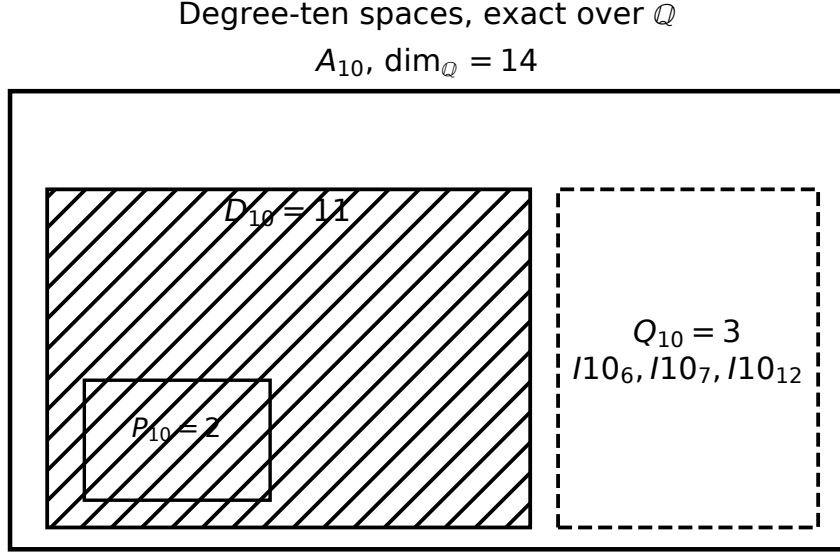


Figure 6. The degree-ten subspaces, exact over $\mathbb{Q}$. $P_{10} \subset D_{10} \subset A_{10}$, and the quotient Q_{10} is spanned by the three free directions $I10_6, I10_7, I10_{12}$.

terms without introducing a Λ -bilinear trace, because any candidate bilinear is again $\langle \Lambda, \Lambda \rangle$ or $\langle V_\Lambda, V_\Lambda \rangle$ and both vanish. What is convention-dependent is the overall normalisation of τ , not the vanishing. A formulation containing a Λ -bilinear that is not a full contraction — for instance one contracted through a background structure other than η — would evade this argument; no formulation in the scope considered here does so.

6 The degree-ten stress flow

6.1 Three spaces that must not be conflated

At degree ten there are several natural subspaces of the invariant space and they have different dimensions. Distinguishing them is the substance of this section; an earlier version of this analysis conflated the first and the fourth and obtained a vacuous answer.

object	dimension	quotient in $\mathcal{A}_{10}$
span of all 37 degree-ten flow targets	14	0
new-forcing space ($\text{Tr}(\tau)$ excluded as a generator)	5	9
static stress span	2	12
$\mathcal{D}_{10}$, the seed closure	11	3

Only the last is $\mathcal{D}_{10}$. The first has no activation condition and therefore reaches everything; the second removes $\text{Tr}(\tau)$ as a generator; the third is static rather than dynamical.

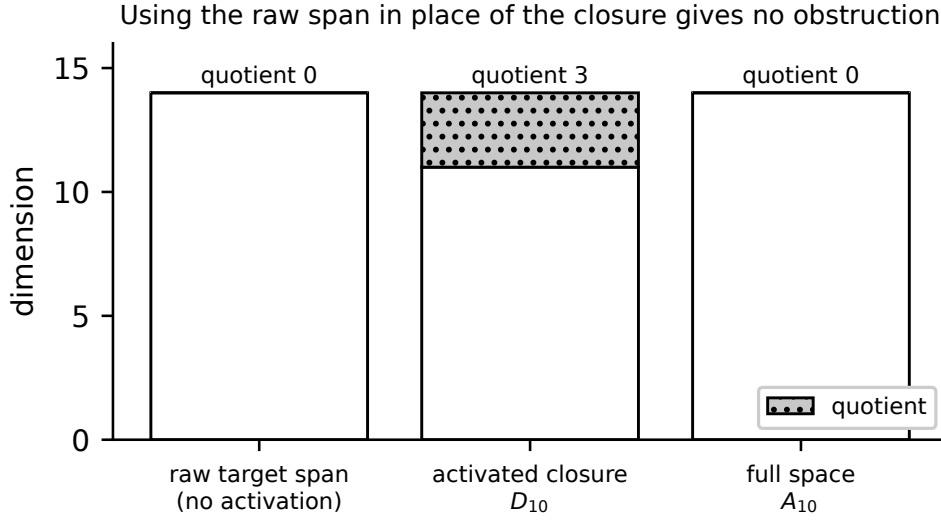


Figure 7. Why the activation rule is the whole content of D_{10} . The raw span of all degree-ten flow targets reaches the full space and leaves no quotient; the activated closure leaves three dimensions.

Definition 6.1 (Activation). A flow target contributes to the closure only once every factor of its coefficient monomial is already reachable. The closure $\mathcal{D}_{10}$ is the fixed point of repeatedly adding targets whose activation condition is met, starting from the seed set.

This is where theorem 5.2 enters. Because $\text{Tr}(\tau)$ has no quadratic part, no degree-two $\text{Tr}(\tau)$ target exists, and nothing in the $\text{Tr}(\tau)$ family is reachable until a degree-four direction is already in the span. Without that, the family would activate immediately.

6.2 The exact result

The flow targets carry exact rational coordinates, so the fixed-point closure can be run over $\mathbb{Q}$ with rational arithmetic rather than modulo a prime. This matters for a reason worth spelling out: $\text{rank}_{\mathbb{F}_p} \leq \text{rank}_{\mathbb{Q}}$, and for a subspace that is *subtracted* this inequality points the wrong way. A $\mathcal{D}_{10}$ that were larger over $\mathbb{Q}$ than modulo p would make the obstruction *smaller*, and agreement across primes could not rule that out. Computing over $\mathbb{Q}$ removes the issue by computing the rank rather than bounding it.

Theorem 6.2 (Degree-ten seed closure). *Over the rationals,*

$$\dim_{\mathbb{Q}} \mathcal{A}_{10} = 14, \quad \dim_{\mathbb{Q}} \mathcal{D}_{10} = 11, \quad \dim_{\mathbb{Q}} \mathcal{Q}_{10} = 3. \quad (6.1)$$

Status: proved, with both bounds certified separately.

The upper bound on $\mathcal{A}_{10}$ needs no computation: the fourteen basis elements *are* the coordinate system, so nothing spanned by them exceeds fourteen, and the modular rank supplies the matching lower bound. For $\mathcal{D}_{10}$ the closure reached its fixed point in three sweeps, order-independently, and its rank was read off by exact rational elimination. The

lower bound is exhibited by an explicit 11×11 minor with determinant

$$12126533474468034240675840 \neq 0, \quad (6.2)$$

computed by two independent routines that agree. The upper bound is established separately by exhibiting three valid annihilators, which imply $\dim \mathcal{D}_{10} \leq 11$ directly. Because the two bounds are obtained by different arguments, their agreement is evidence and not a tautology.

The three basis directions the seeded flow does not reach are $I_{10,6}$, $I_{10,7}$ and $I_{10,12}$, corresponding to free columns 5, 6 and 11; the modular record agrees with the rational computation on both the rank and the identity of the free columns.

Remark 6.3 (Scope). $\mathcal{D}_{10}$ is a *seed* closure. The generator-extension problem $dV/d\lambda = f(T, S, \lambda)$, in which additional generators are permitted, is a different question and is not answered here. Nothing in theorem 6.2 should be read as addressing it.

6.3 The obstruction is load-bearing

A rule that nothing can violate is not being enforced. To check that the activation semantics are doing real work, we insert the degree-two trace contribution that theorem 5.2 forbids — a quadratic trace is unconditional, so its coefficient monomial is empty and the $\text{Tr}(\tau)$ family activates immediately — and recompute:

quantity	theorem 5.2 holds	quadratic trace inserted
$\dim \mathcal{D}_{10}$	11	14
$\dim \mathcal{Q}_{10}$	3	0

This is not a perturbation of the answer. It is the difference between an obstruction and no obstruction, and it reproduces exactly the error made in an earlier stage of this work, where taking the unconditional span of all raw targets gave rank 14 and quotient 0 with nothing objecting at the time. If the free stress tensor were not traceless, there would be no degree-ten obstruction to report.

6.4 Independent confirmation from the rational flow system

The coefficient system for $dV/d\lambda = f(T, \lambda)$ was rationally reconstructed through degree twelve, fit on five primes and validated on an independent holdout. Its new-forcing span — what the flow can create with the homogeneity generator $\text{Tr}(\tau)$ excluded — is:

degree	full	new forcing	obstruction
4	1	1	0
6	2	1	1
8	7	3	4
10	14	5	9
12	72	21	51

This uses a different script, a different algorithm, and rational rather than modular arithmetic, and it recovers the same one-dimensional sextic obstruction found independently at degree six. We record it as a cross-check, not as a second proof of theorem 6.2: the quantity tabulated here is the new-forcing span, not $\mathcal{D}_{10}$, and the two must not be identified.

7 What is not established

We regard this section as part of the result rather than an appendix to it.

7.1 $\text{Tr}(M^6)$ has no certified rational lift

At degree twelve, the stress row associated with $\text{Tr}(M^6)$ was carried through CRT lifting at fifteen primes, giving a modulus of approximately 5.2×10^{67} . Twenty-nine of seventy-two columns still exceed the uniqueness bound for rational reconstruction. The coefficients were therefore *not* guessed; $\text{Tr}(M^6)$ was adopted as a stress-adapted basis element instead.

The contrast with the flow coefficients is the informative part. Those lifted cleanly at five primes, while this single degree-twelve row remains unlifted at fifteen. That pattern points to a genuine height problem specific to $\text{Tr}(M^6)$ rather than to insufficient computation, which is why the natural route forward is an analytic identity rather than more primes.

The four-dimensional *rank* at this degree is certified. The *coordinates* are not, and no rational $\text{Tr}(M^6)$ coordinates appear anywhere in this paper.

7.2 Spaces certified only over the tested fields

space	value	status
$\mathcal{A}_{10}$	14	proved over $\mathbb{Q}$
$\mathcal{G}_{10}$	12	proved over $\mathbb{Q}$
$\mathcal{P}_{10}$	2	proved over $\mathbb{Q}$
$\mathcal{D}_{10}$	11	proved over $\mathbb{Q}$
$\mathcal{Q}_{10}$	3	proved over $\mathbb{Q}$
$\mathcal{B}_{10}$	12	verified over the tested fields only
$\mathcal{B}_{10} \cap \mathcal{P}_{10}$	1	verified over the tested fields only

$\mathcal{A}_{10}$, $\mathcal{G}_{10}$ and $\mathcal{P}_{10}$ are spans of distinct standard basis vectors in atlas coordinates — all fourteen, the twelve graph generators, and the two lower products respectively. A set of k distinct unit vectors has rank k over any field, so their dimensions are structural and no prime can be exceptional.

$\mathcal{B}_{10}$ is different. It is the span of the published equation-(4.24) candidates, and each candidate's coordinate vector is *recovered* by solving a linear system against a design matrix modulo p , so the coefficients are modular objects. We therefore have $\dim_{\mathbb{Q}} \mathcal{B}_{10} \geq 12$ from the modular rank and $\dim_{\mathbb{Q}} \mathcal{B}_{10} \leq 14$ from $\mathcal{B}_{10} \subseteq \mathcal{A}_{10}$, and nothing between them is established. The intersection $\mathcal{B}_{10} \cap \mathcal{P}_{10}$ inherits the gap through $\dim(B \cap P) = \dim B +$

$\dim P - \dim(B + P)$. Its modular value is 1, consistent across the tested primes, and that is exactly what we state.

Closing this would require a rational evaluator for the atlas elements, which does not exist in the present package. The route is not exotic — the candidates’ modular images at four primes give a CRT modulus of roughly 10^{18} , so a rational lift is unique when numerator and denominator are each below about 10^9 — but whether the entries are that small is not known, and it was not attempted here.

7.3 Degree twelve

Degree-twelve data is a certified *input* to theorem 4.1, contributing 60 of the 81. It does not enter the title, the abstract or the central claims of this paper, because doing so would require a complete atlas, a verified product/primitive split, an exact rank certificate, a trace/spinor comparison and holdout validation at that degree, and not all of those are in hand. The scope decision is deliberate and is recorded as such.

7.4 Claims we do not make

We do not derive a correction to the type IIB effective action, and nothing here should be read as one. We make no claim about causality, hyperbolicity, supersymmetric completion or positivity of any theory in the family, because none of those was calculated. We do not claim the enumerate–evaluate–relate workflow, which is due to [5]. Novelty relative to the existing literature is not asserted anywhere in this paper; a systematic comparison against the relevant primary sources has not yet been carried out, and until it has, no priority claim would be responsible.

8 Discussion

The degree-ten quotient $\dim_{\mathbb{Q}} \mathcal{Q}_{10} = 3$ says that a seeded stress-tensor flow in the family of [2] does not reach every degree-ten invariant of a self-dual five-form. Three directions — $I_{10,6}$, $I_{10,7}$, $I_{10,12}$ — lie outside the closure. Whether they are reachable once additional generators are admitted is the generator-extension problem, and we have been careful not to answer it by accident.

The tracelessness result of section 5 is the kind of statement that is easy to assume and cheap to prove once asked, and the counterfactual in section 6.3 shows the cost of assuming it wrongly. We would flag it as the natural first thing to check when this analysis is extended to the chiral two-form in $D = 6$, where the analogue of lemma 5.1 depends on $p = 3$ being odd and so goes through, and to $T\bar{T}$ -like flows in higher dimensions more generally [11, 12].

Two questions we would put to a reader with more context than we have. First: is there a known analytic identity expressing $\text{Tr}(M^6)$ in terms of the atlas generators? A single such identity removes the largest limitation recorded in section 7.1. Second: equation (2.36) of arXiv:2509.14351v2 prints a trace sign that appears inconsistent with (2.33) and with the reproduced $V(I_4)$ formula; we have not been able to determine which is intended, and our implementation follows (2.33).

Author contributions

We state contributions explicitly because the credit situation here is not the default one and would be easy to misread.

The enumerate–evaluate–detect–relations workflow used in section 4 is due to C.F. and collaborators [5] and predates this work; it is used here, not introduced. U.S.K. posed the degree-ten closure question, carried out the exact-arithmetic execution of that workflow for the five-form problem, produced the rank certificate, proved the tracelessness lemma of section 5, and computed the degree-ten seed closure over $\mathbb{Q}$. The corresponding author is not yet designated; that is one of the decisions this draft is circulated to settle. U.S.K. is responsible for the submission and for correspondence with referees.

The count of 81 functionally independent invariants, and the degree-ten candidate list, are due to the source literature [1]; the nonlinear chiral four-form family and its stress flow are due to [2]. This paper contributes certificates, a proof, and an exact closure computation on top of those.

Acknowledgments

This work made substantial use of AI assistance. The extent is documented in `manuscript/jhep/ai_assista` and is summarised here: most of the computational code, all of the test suites, and much of the audit apparatus were machine-written, as was the analysis that produced the exact degree-ten result. No AI system is an author of this paper. The disclosure document also records what this assistance does *not* amount to, in particular that it does not constitute independent human verification of the results.

A The frame orientation, and how it was found

The bridge of section 3 depends on a change of frame L satisfying $L^\top \eta_{\text{null}} L = \eta_{\text{Lor}}$. That congruence is solved modulo p , and the solve takes a modular square root. A square root modulo p has two values, r and $p - r$; both give a valid congruence, and they differ by a reversal of the frame’s orientation.

Orientation is not cosmetic here. Reversing it reverses the sign of the volume form, hence of the Hodge star, hence exchanges the star’s $+1$ and -1 eigenspaces — which is to say it exchanges which of the self-dual and anti-self-dual five-forms the gamma map annihilates. Left unpinned, the bridge realises the self-dual projector at some primes and the anti-self-dual projector at others, with nothing to announce the difference until a later step reports an image of dimension zero.

How it surfaced. It surfaced through a holdout prime, which is what holdout primes are for. Nine fitting cells and three holdout cells at $p = 32713$ completed normally; all three cells at $p = 32707$ failed identically, with the left-inverse construction reporting an image of dimension 0 instead of 126. The bridge validation suite could not have caught this: it exercises two primes, both of which happen to receive the orientation-preserving branch.

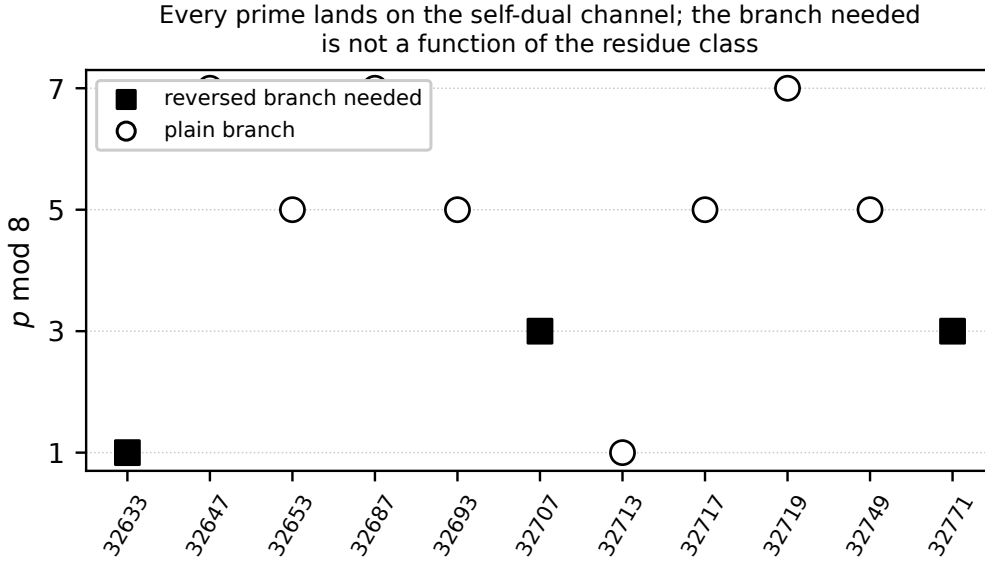


Figure 8. The orientation-normalised frame at 11 primes across all four usable residue classes modulo 8. Filled squares mark the primes needing the reversed square-root branch: 32633, 32707 and 32771, in classes 1, 3 and 3. Every prime lands on the self-dual channel.

A wrong diagnosis, and why it was wrong. The first explanation was arithmetic: 32707 is 3 mod 8, the other tested primes were not, and a second failing prime 32771 is also 3 mod 8. On two examples the pattern looked like a congruence condition, and 32707 was replaced by 32693 as a holdout while the matter was open.

That diagnosis is false. Once the orientation is pinned, $p = 32633$ — which is 1 mod 8 — requires exactly the same branch as 32707 and 32771. A prime outside the suspected class needs the identical correction, so the branch requirement was never a function of the residue class. It was a coincidence of the only two examples available when the rule was written.

The fix. The construction now selects the branch under which the self-dual space survives, in a fixed order, and this is a property of the construction rather than a per-prime exception. Every residue class then works. Table 1 records eleven primes across all four usable classes modulo 8; all eleven land on the self-dual channel, and the three that need the reversed branch are 32633, 32707 and 32771 — classes 1, 3 and 3.

Consequences. Three, and the third is the reason this section exists rather than a footnote.

First, 32707 is restored to the holdout set and produces ordinary cells at rank 81. Nothing about it is exceptional. Second, 32693 is retained as an *additional* validation prime rather than a substitute, and the assembly step refuses a matrix that carries the substitute while missing the original, since such a matrix has the right number of cells and passes every per-cell check.

prime	$p \bmod 8$	branch	channel	left inverse	congruence
32633	1	reversed	selfdual	yes	holds
32647	7	plain	selfdual	yes	holds
32653	5	plain	selfdual	yes	holds
32687	7	plain	selfdual	yes	holds
32693	5	plain	selfdual	yes	holds
32707	3	reversed	selfdual	yes	holds
32713	1	plain	selfdual	yes	holds
32717	5	plain	selfdual	yes	holds
32719	7	plain	selfdual	yes	holds
32749	5	plain	selfdual	yes	holds
32771	3	reversed	selfdual	yes	holds

Table 1. The orientation-normalised frame at eleven primes covering all four usable residue classes modulo 8. Every prime lands on the self-dual channel. The three needing the reversed square-root branch are 32633, 32707 and 32771, in classes 1, 3 and 3 respectively — so the branch requirement is not a function of the residue class. Verified by `scripts/verify_orientation_canonical_independent.py`, which does not call the production selector.

Third, every certificate in this paper was regenerated after the fix. The recomputation was not expected to change any value, and it did not: at each prime where the plain branch already survived, the pinned frame is bit-identical to the unpinned one, and cells recomputed under the corrected code agree with their predecessors in every candidate value, output hash, Jacobian block and Euler check. The earlier numbers were right. They were regenerated because a certificate should carry one evaluator’s provenance rather than two, which is a different requirement from being correct.

B Computational methodology

B.1 One writer per artifact

Every scientific artifact in this work is produced by exactly one process. Each computational cell runs in its own process, writes its own file, takes a lock, and completes with an atomic rename. Each carries a unique execution identifier, the hash of the frozen source it ran against, and hashes of the candidate list and coordinate ordering it used. A cell that is interrupted leaves a record saying so; a cell that completes leaves a terminal-status record. Resumption skips completed cells rather than recomputing or, worse, appending to them.

This discipline is not decorative. An earlier stage of this work ran six concurrent test processes writing a single log, and the resulting file could not be used as evidence for anything.

B.2 Exact arithmetic throughout

No result in this paper depends on a floating-point comparison or a tolerance. Modular computations are exact in $\mathbb{F}_p$; rational computations use exact rational arithmetic; determinants are computed by fraction-free elimination [4] and cross-checked by Laplace expansion.

prime	role	seed	rank	rows	Euler	errors	zero rows
32717	fitting	11	81	83	83/83	0	0
32717	fitting	22	81	83	83/83	0	0
32717	fitting	33	81	83	83/83	0	0
32719	fitting	11	81	83	83/83	0	0
32719	fitting	22	81	83	83/83	0	0
32719	fitting	33	81	83	83/83	0	0
32749	fitting	11	81	83	83/83	0	0
32749	fitting	22	81	83	83/83	0	0
32749	fitting	33	81	83	83/83	0	0
32707	holdout	11	81	83	83/83	0	0
32707	holdout	22	81	83	83/83	0	0
32707	holdout	33	81	83	83/83	0	0
32713	holdout	11	81	83	83/83	0	0
32713	holdout	22	81	83	83/83	0	0
32713	holdout	33	81	83	83/83	0	0
32693	extra	11	81	83	83/83	0	0
32693	extra	22	81	83	83/83	0	0
32693	extra	33	81	83	83/83	0	0

Table 2. The complete certificate matrix under the canonical orientation-fixed bridge: fifteen required publication cells and three extra validation cells at 32693. Every cell evaluates all 83 candidates with no errors and no zero rows.

prime	modular LU	Bareiss	agree	nonzero
32693	9965	9965	yes	yes
32707	29206	29206	yes	yes
32713	12844	12844	yes	yes
32717	9699	9699	yes	yes
32719	31196	31196	yes	yes
32749	20755	20755	yes	yes

Table 3. The same explicit 81×81 minor of the integral Jacobian, evaluated at six primes by two independent determinant routines. A minor that is nonzero modulo any prime is nonzero over $\mathbb{Z}$, so $\text{rank}_{\mathbb{Q}} \geq 81$ unconditionally.

Where a rational lift from modular data is performed, it is validated on held-out primes that were not used in the fit, and a lift that fails validation is reported as a failure rather than retried until it passes.

B.3 No stopping rules from published values

No enumeration or rank computation in this work terminates because it reached a published target. The values 81, 14, 11 and 3 appear in the literature and are used here for validation only. Every cell evaluates its full candidate list and reports whatever rank it finds.

space	modular	exact over $\mathbb{Q}$	status
A_{10}	14	14	PROVED
G_{10}	12	12	PROVED
P_{10}	2	2	PROVED
D_{10}	11	11	PROVED
Q_{10}	3	3	PROVED
B_{10}	12	<i>not established</i>	CERTIFIED AT THE TESTED PRIMES
$B_{10} \cap P_{10}$	1	<i>not established</i>	at the tested primes

Table 4. Degree-ten subspaces. A_{10} , G_{10} and P_{10} are spans of distinct basis vectors, so their dimensions are structural and no prime can be exceptional. D_{10} and Q_{10} are established over $\mathbb{Q}$ by exact rational computation. B_{10} is recovered by a solve modulo p and keeps its finite-field qualification.

suite	collected	passed	failed	errors	exit
tensor	290	290	0	0	0
bridge	176	176	0	0	0
total		466	0		

Table 5. Test suites, counted by a parser with nineteen regression tests of its own covering wrapped progress output, interrupted runs and duplicate-writer contamination. A log without a recorded exit status is reported as incomplete rather than as a pass.

C Certificate index

Each claim in the body is backed by a machine-readable artifact and by tests that guard it. The correspondence is:

claim	artifact	guard
prop. 3.1	gate 1.2 record	test_bridge.py
thm. 4.1	results/rank81/	test_matrix_aggregator.py
thm. 5.2	g10_trace_verification.json	test_g10_trace_activation.py
prop. 5.3	interacting_flow_equations.json	test_g10_trace_activation.py
thm. 6.2	D10_exact_rational_final.json	test_stress_flow_closure.py
§6.3	counterfactual record	test_g10_trace_activation.py

D Conventions, and how they compare to the literature

Sign conventions are the usual source of irreproducibility in this area, so we record ours against the sources rather than leaving them implicit.

quantity	this paper	note
signature	$(-, +, \dots, +)$	mostly plus
$\varepsilon^{01\dots 9}$	-1	fixes the sign of $\star$
duality	$\star\Lambda = \Lambda$	Λ self-dual
V_Λ duality	anti-self-dual	structural, used in prop. 5.3
Λ on shell	$\Lambda_5 = -F_5^+$	identification of [2]
degree	polynomial degree in Λ	<i>not</i> coupling order
τ	$48T$	the M -normalised tensor

Nothing in section 5 depends on the choice of $\varepsilon^{01\dots 9}$: the independent verifier tests both signs and both duality channels and finds the trace vanishing in all of them. The 48 in the definition of τ is a normalisation and propagates into the coefficient $10(d-2)$; a different normalisation rescales that coefficient without moving its zero, which is at $d=2$ regardless.

One discrepancy in the sources should be flagged. Equation (2.36) of arXiv:2509.14351v2 prints a trace sign that we were unable to reconcile with (2.33) of the same paper or with the reproduced $V(I_4)$ formula. Our implementation follows (2.33). Readers reproducing this work who follow (2.36) instead should expect a sign mismatch at that step, and we would welcome clarification.

E Reproduction

The computational record is designed to be re-run rather than trusted.

What a reproduction has to produce. Re-running the rank certificate should yield, for every (prime, seed) cell independently: 83 candidates scheduled, 83 evaluated, zero errors, zero interrupted, zero skipped, Euler homogeneity 83/83, no zero rows, coordinate dimension 126, rank 81, and cumulative rank by degree (1, 3, 9, 21, 81) at degrees (4, 6, 8, 10, 12). A cell that disagrees with any of these is a failed reproduction and should be reported as one; none of these numbers is a tolerance.

Fitting and holdout. The prime matrix separates fitting primes from holdout primes. Rational lifts are fit on the former and validated on the latter, and a lift that fails holdout validation is discarded rather than refit until it passes. This matters most for the flow coefficient system of section 6, where the lift succeeded, and for $\text{Tr}(M^6)$ (section 7.1), where it did not and is reported as not having.

Choice of primes. Every prime used must satisfy $p \not\equiv 3 \pmod{8}$, for the reason given in section 4.4. A cell at such a prime does not produce a wrong answer; it fails to produce one at all, which is the intended behaviour.

Determinism. Sample points are drawn from seeded generators, and the seed is recorded in each cell’s output alongside the hash of the frozen source, the candidate-list hash and the coordinate-ordering hash. Two runs with the same seed and the same source hash must agree bit for bit; two runs with different seeds must agree on every rank.

Independence of the degree-ten result. Theorem 6.2 is reproduced by a verifier that shares no code with the production closure: it recomputes the raw target span, the activated closure and the complete invariant space from the flow artifact directly, checks order-independence across sweeps, and certifies the lower and upper bounds by different arguments (an explicit minor, and a set of annihilators). Agreement between the production path and this verifier is the check we would want a referee to run first.

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